

THE FIFTEENTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 5, 1955

Morning Session

1. Prove that there is no set of integers m, n, p except $0, 0, 0$ for which $m + n\sqrt{2} + p\sqrt{3} = 0$. (page 402)

2. $A_1 A_2 \dots A_n$ is a regular polygon inscribed in a circle of radius r and center O . P is a point on line OA_1 extended beyond A_1 . Show that

$$\prod_{i=1}^n \overline{PA_i} = \overline{OP}^n - r^n. \quad (\text{page 403})$$

3. Suppose that $\sum_{i=1}^{\infty} x_i$ is a convergent series of positive terms which monotonically decrease (that is, $x_1 \geq x_2 \geq x_3 \geq \dots$). Let P denote the set of all numbers which are sums of some (finite or infinite) subseries of $\sum_{i=1}^{\infty} x_i$. Show that P is an interval if and only if

$$x_n \leq \sum_{i=n+1}^{\infty} x_i \quad \text{for every integer } n. \quad (\text{page 403})$$

4. On a circle, n points are selected and the chords joining them in pairs are drawn. Assuming that no three of these chords are concurrent (except at the endpoints), how many points of intersection are there? (page 405)

5. If a parabola is given in the plane, find a geometric construction (ruler and compass) for the focus. (page 405)

6. Find a necessary and sufficient condition on the positive integer n that the equation

$$x^n + (2 + x)^n + (2 - x)^n = 0$$

have a rational root. (page 406)

7. Consider the function f defined by the differential equation

$$f''(x) = (x^3 + ax)f(x)$$

and the initial conditions $f(0) = 1, f'(0) = 0$. Prove that the roots of f are bounded above but unbounded below. (page 407)

Afternoon Session

1. A sphere rolls along two intersecting straight lines. Find the locus of its center. (page 408)

2. Suppose that f is a function with two continuous derivatives and $f(0) = 0$. Prove that the function g , defined by $g(0) = f'(0)$, $g(x) = f(x)/x$ for $x \neq 0$, has a continuous derivative. (page 410)

3. Prove that there exists no distance-preserving map of a spherical cap into the plane. (Distances on the sphere are to be measured along great circles on the surface.) (page 411)

4. Do there exist 1,000,000 consecutive integers each of which contains a repeated prime factor? (page 412)

5. Given an infinite sequence of 0's and 1's and a fixed integer k , suppose that there are no more than k distinct blocks of k consecutive terms. Show that the sequence is eventually periodic. (For example, the sequence 11011010101 followed by alternating 0's and 1's indefinitely, which is periodic beginning with the fifth term.) (page 413)

6. Prove: If $f(x) > 0$ for all x and $f(x) \rightarrow 0$ as $x \rightarrow \infty$, then there exists at most a finite number of solutions of

$$f(m) + f(n) + f(p) = 1$$

in positive integers m, n , and p . (page 414)

7. Four forces acting on a body are in equilibrium. Prove that, if their lines of action are mutually skew, they are rulings of a hyperboloid. (page 415)